

Inversion of GPS occultation measurements using Fresnel diffraction theory

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Abstract. Occultation data from the GPS/MET experiment are inverted to temperature profiles of the neutral atmosphere of the Earth using the Fresnel inversion technique. The technique is not limited in resolution by diffraction effects thus a good vertical resolution is achieved. In the derivation a thin screen approximation is used. The influence of this approximation on the results is discussed. The results obtained using the Fresnel inversion is compared to results obtained using the traditional geometrical optics inversion approach and to numerical weather prediction data.

1. Introduction

Promising results of inversions of the radio occultation measurements of the Earth's atmosphere performed in the GPS/MET experiment with the American satellite Microlab-1 have been published [Ware *et al.*, 1996], [Kursinski *et al.*, 1996]. The results have been obtained using a geometrical optics approach and the Abel transform [Melbourne *et al.*, 1994]. This method gives a very good vertical resolution - varying from a few hundred meters close to the Earth to the order of 1km at higher altitudes. It is well known though that the resolution could be further improved if diffraction effects could be taken into account. Thus, there has recently been much effort in developing inversion methods that take into account the diffraction effects [Mortensen and Høeg, 1998], [Melbourne *et al.*, 1994], [Karayel and Hinson, 1996], [Gorbunov *et al.*, 1996].

The Fresnel inversion in [Melbourne *et al.*, 1994] and [Mortensen and Høeg, 1998] is based on using a thin screen as an approximation for the Earth's atmosphere. The method has been adapted from the planetary missions where it has been used with success [Marouf *et al.*, 1986], [Hinson *et al.*, 1997]. Thereby the atmospheric influence on the signal propagation can be isolated using Fresnel diffraction theory and a vertical atmospheric profile can be derived. This profile will be without the

usual diffraction effects but as the atmosphere in general contains 3D inhomogeneities which influences the signal the derived profile will contain high-frequency artifacts. The thin screen approximation poses a problem for the dense part of the atmosphere of the Earth, however, in [Mortensen and Høeg, 1998] it has been demonstrated that the inversion can be formulated such that this problem is of limited influence on the results.

So far, only simulated occultations for the Earth have been inverted using the Fresnel inversion. This paper briefly reviews the Fresnel inversion and presents the work done on adjusting the technique to invert real occultation data for the Earth taking noise into account. The new inversion results are discussed in comparison to the geometrical optics inversion and numerical weather prediction data.

2. The Fresnel Inversion

The occultation geometry is shown in Figure 1. In this figure, GPS denotes one of the GPS satellites, and LEO refers to the low Earth orbiting satellite which measures the signals from the GPS while the LEO gets occulted relative to the GPS. A two dimensional approximation is used and thus, a rectangular xh -coordinate system can be introduced. The x -axis is parallel to the direction from the GPS to the LEO. The origin of the coordinate system is at the limb of the Earth. At $x = 0$, a thin screen S is placed. This thin screen represents the influence from the atmosphere of the Earth on the signal propagation. A ray path for the propagation of a signal is shown in the figure. The signal propagates in free space to the thin screen. In the atmosphere, the signal is delayed and the direction is changed by the

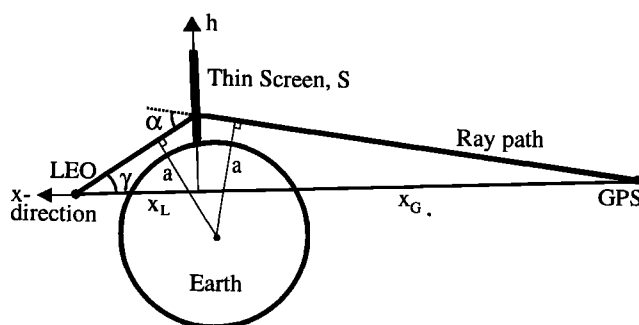


Figure 1. Occultation geometry for the thin screen approximation.

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angle α relative to the free space propagation. After having passed the thin screen, the signal propagates in free space to the LEO. The quantities x_L and x_G are distances from the thin screen to the LEO and to the GPS, respectively. The angle γ is the angle between the ray path at the LEO and the x-axis. The right-angled distance from the center of the Earth to the ray path is given by a . All the parameters shown in Figure 1 can be found from the measured phase data and the geometry of the occultation using the ray optical approach [Melbourne *et al.*, 1994]. For the described geometry and assuming an $\exp(-i\omega t)$ time dependence the Helmholtz-Kirchoff integral theorem can be written as [Mortensen and Høeg, 1998]

$$\bar{E}(h_b) \exp[i\Delta\phi(h_b)] = \frac{1-i}{\sqrt{2D\lambda}} \int_{-\infty}^{\infty} A(h) \exp \left[i \left(\frac{\pi (h-h_b)^2}{\lambda D} + \Psi(h) \right) \right] dh \quad (1)$$

where $\bar{E}(h_b)$ and $\Delta\phi(h_b)$ are the amplitude and phase, respectively, of the signal measured at the LEO relative to the signal measured at the LEO in the absence of the atmosphere and the Earth. Their functional argument h_b is the altitude of the LEO-GPS line at the point of intersection with the thin screen. Using the assumption of a tenuous atmosphere the altitude h_b can be found from $h_b = (a - R_e) - x_L \gamma$ where R_e is the radius of the Earth. In (1), $A(h)$ and $\Psi(h)$ are the amplitude and phase delay, respectively, due to the atmosphere. Furthermore, λ is the wavelength and D is the reduced distance, $\frac{1}{D} = \frac{1}{x_L} + \frac{1}{|x_G|}$.

In a form slightly modified to account for the extent of the atmosphere and the presence of noise in the measured signal, the inverse transform of (1) - the Fresnel transform - is given by [Mortensen and Høeg, 1998]

$$A(v) \exp[i\Psi(v)] = \frac{1}{1-i} \int_{-\infty}^{\infty} \bar{E}(u_b) \exp[i\Delta\phi(u_b)] w(u_b - v) \exp \left[-\frac{i\pi}{2} (u_b - v)^2 \right] du_b \quad (2)$$

where $u_b = h_b \sqrt{2/\lambda D(h)}$, $v = h \sqrt{2/\lambda D(h)}$, and w is a weighting function $w(u_b - v) = \cos^2[\pi(u_b - v)/W]$ for $|u_b - v| \leq W/2$ and $w(u_b - v) = 0$ otherwise. The width W determines the integration interval. The differences to a direct inversion of (1) is that D now depends on altitude and that the weighting function w has been introduced.

The variation of $D(h)$ has been determined as follows: For each sample in a measurement, a temporary value $D_s(h)$ is determined using the instantaneous positions of the satellites and the calculated geometrical parameters. An upper and lower limit for $D(h)$ is obtained as the average value of $D_s(h)$ for $40 \leq h \leq 100\text{km}$ and $0 \leq h \leq 7.5\text{km}$, respectively. Values of $D(h)$ in between the upper and lower limit are then found by interpolation. This empirically change in the inversion gives good results because a varying distance $D(h)$

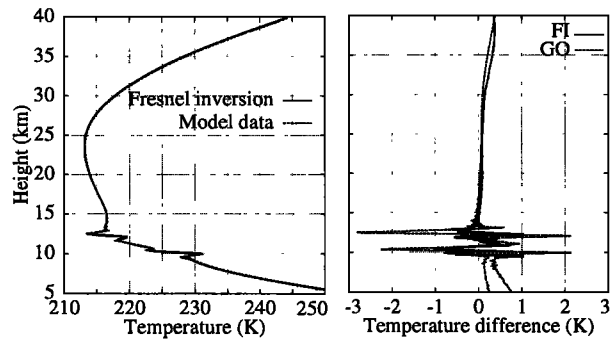


Figure 2. Temperature profile obtained with the Fresnel inversion compared to the input model data. The panel on the right shows the temperature difference to the model data for the Fresnel inversion (FI) and the geometrical optics inversion (GO).

complies better with the actual situation in which the satellites move and the atmosphere has an extent.

Use of the weighting function w has the effect of smoothing the signal [Marouf *et al.*, 1986]. The width W has been chosen variable in order to properly account for both the noise in the signal which requires a small integral interval and for the fast varying phase close to the Earth which requires large integration in-

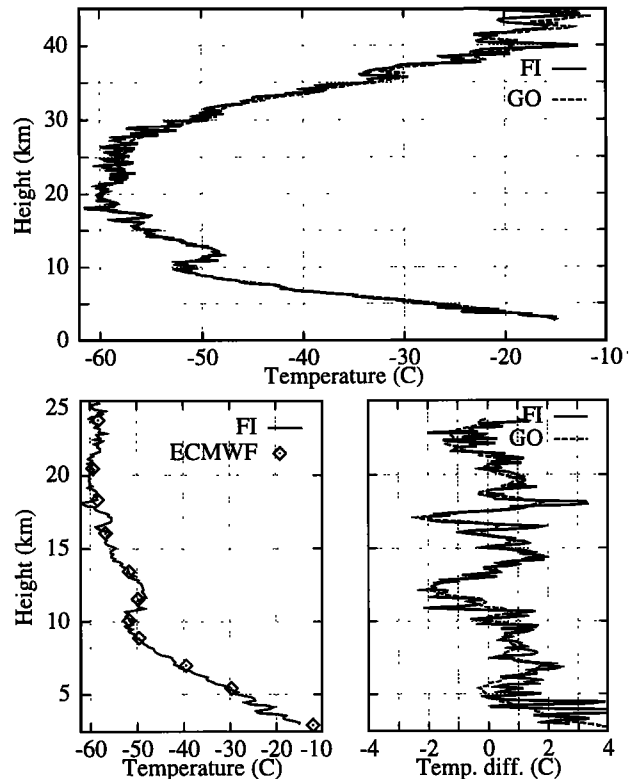


Figure 3. Upper panel: Temperature results of Fresnel inversion and geometrical optics inversion of GPS/MET occultation no. 256, Feb. 4, 1997. Lower left panel: Fresnel inversion temperature result shown together with ECMWF data points. Lower right panel: Temperature difference to ECMWF data for Fresnel inversion (FI) and geometrical optics inversion (GO).

tervals for the integral to converge. That is, $W = 3$ for $h \geq 45\text{km}$ and $W = 6$ for $45\text{km} > h \geq 30\text{km}$ and $W = 600 \exp(-0.15h)$ for $h < 30\text{km}$.

The Fresnel transform (2) shows that it is possible to transform the measured complex signals to attenuation and phase delay due to the atmosphere. The GPS signals are transmitted and measured at two frequencies f_1 and f_2 . The Fresnel transform (2) is performed for both signals and subsequently, the ionosphere correction is performed as

$$\varphi(h) = \frac{c}{2\pi} \frac{f_2 \Psi_2(h) - f_1 \Psi_1(h)}{f_2^2 - f_1^2} \quad (3)$$

Using the ionosphere corrected phase $\varphi(h)$ the refractive index can be found as

$$n(a) = \exp \left(-\frac{1}{\pi} \frac{d}{da} \left[\int_a^\infty \frac{a\varphi(\xi - R_e)}{\xi \sqrt{\xi^2 - a^2}} d\xi \right] \right) \quad (4)$$

when assuming spherical symmetry and a tenuous atmosphere. The altitude corresponding to a is $h_a = (a/n(a)) - R_e$. The temperature can be retrieved from the refractivity as usual [Kursinski et al., 1996].

3. Inversion Results

The first results shown are inversions of a simulated occultation. The atmosphere model is a dry air climatological model with a spherically symmetric disturbance superposed in the upper troposphere. The forward occultation simulation has been performed with a 3D ray tracing program [Høeg et al., 1996] adding the complex fields from multiple rays in the region around the disturbance. This represents an approximate solution to the forward problem [Gorbunov et al., 1996]. No ionosphere model has been used and a spherical Earth was used. The simulation was based on orbit ephemerides from GPS/MET occultation no. 43, September 22, 1995 (52.0°N, 23.0°E).

Figure 2 shows the model temperature profile compared with the temperature profile obtained with the Fresnel inversion. The panel on the right shows the temperature difference between the Fresnel inversion and the model data as well as the difference between the geometrical optics inversion and the model data. As can be seen from the figure, oscillations occur in the error of the retrieved temperature for both methods around the disturbance. A part of these oscillations are likely to be due to the inaccuracies in the forward modeling. The disturbance is reasonably well resolved by the Fresnel inversion with errors below $\pm 1\text{K}$ overall. The geometrical optics inversion has significantly larger errors - up to -2.8K - illustrating the problems of this method in resolving large gradients due to the limited vertical resolution. As can also be seen from the figure, the error of the Fresnel inversion increases close to the Earth. Below approximately 3km the error is larger than 1K and the results cannot any longer be said to have the sufficient

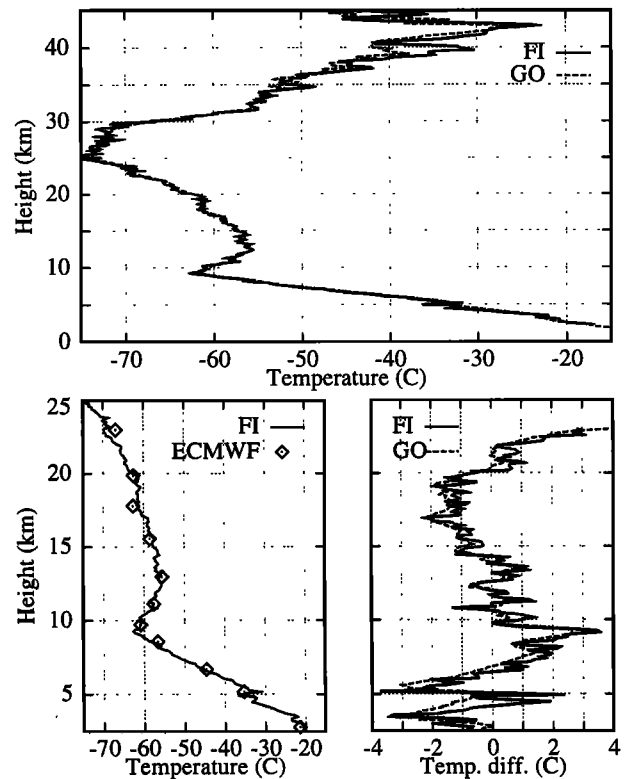


Figure 4. Upper panel: Temperature results of Fresnel inversion and geometrical optics inversion of GPS/MET occultation no. 70, Feb. 4, 1997. Lower left panel: Fresnel inversion temperature result shown together with ECMWF data points. Lower right panel: Temperature difference to ECMWF data for Fresnel inversion (FI) and geometrical optics inversion (GO).

accuracy. This is due to the tenuous atmosphere assumption used in the thin screen approximation which is not valid in the densest part of the atmosphere close to the Earth [Mortensen and Høeg, 1998].

The next results are inversions of real measured data. Figure 3 shows inversion results of GPS/MET occultation no. 256, February 4, 1997. This occultation took place at 52.1°N, 6.2°W. The GPS anti-spoofing system was turned off. The upper panel of the figure shows the result from the Fresnel inversion and the geometrical optics inversion. In general, the two results are very similar. As expected, the Fresnel inversion has more variations showing bigger gradients than the geometrical optics solution but both solutions have the same structure. Parts of the high-frequency oscillations seen in the Fresnel inversion are due to derivations from spherical symmetry in the atmosphere. That is, the results seen is in general a signature of 3D-inhomogeneities. Particularly above 20km these oscillations have a significant amplitude. In the left lower panel of the figure the Fresnel inversion result is shown along with numerical weather prediction data from the European Center for Medium-Range Weather Forecasting (ECMWF). The ECMWF data compare with the occultation inversions where water vapour is ignored because the chosen northern lati-

tude winter data have a low water vapour content. The lower right panel of the figure shows the difference between the Fresnel inversion data and the ECMWF data and the geometrical optics data and the ECMWF data. A linear interpolation has been performed between the points in the ECMWF data. In most areas, the differences to the ECMWF data are less than $\pm 2^\circ\text{C}$ for both inversion methods. For both methods, the difference to the ECMWF data increases close to the Earth. The Fresnel inversion gives a somewhat larger difference close to the Earth as expected from the results shown in Figure 2. Above 5 km it seems likely that the major part of the difference to the ECMWF data for both inversions are due to the lack of vertical resolution in the ECMWF data.

Figure 4 shows another example of inversion results. The data are from GPS/MET occultation no. 70, February 4, 1997. This occultation took place at 73.3°N , 148.0°W . Again, the upper panel of the figure shows the Fresnel inversion and the geometrical optics inversion results together. The results compare well having the same type of differences as the data shown in Figure 3. In the lower left panel of the figure, the Fresnel inversion and the ECMWF data are shown together and on the right, the differences from the inversion results to the ECMWF data are shown. Again, both methods produces differences less than $\pm 2^\circ\text{C}$ in most areas. As in Figure 3 the differences to the ECMWF data has a tendency to increase somewhat for both methods for altitudes less than 5 km. This time, the Fresnel inversion giving the same type of results as the geometrical optics inversion. That is, the tenuous atmosphere assumption in the Fresnel inversion does not seem to be a problem in this case. The reason for this is the gradient in the data around 5 km which the two methods resolves differently, thereby changing the overall retrieval results in the area.

4. Conclusions

Results of inversion of measured radio occultation data for the Earth using the Fresnel inversion has been shown for the first time. In comparison to the geometrical optics inversion method more structure is seen in the Fresnel inverted results. Inversion of simulated occultations suggests that the Fresnel inversion resolves the large temperature gradients whereas the geometrical optics solution gives a more averaged value of the true result. However, horizontal inhomogeneities introduces high frequency artifacts in the results. Further-

more, the accuracy of the Fresnel inversion for altitudes below approximately 5 km could be better.

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References

- Gorbunov, M. E., A. S. Gurvich and L. Bengtsson, Advanced algorithms of inversion of GPS/MET satellite data and their application to reconstruction of temperature and humidity, *Rep. 211*, Max-Planck-Institute for Meteorology, 1996.
- Hinson, D. P., F. M. Flasar, A. J. Kliore, P. J. Schinder, J. D. Twicken and R. G. Herrera, Jupiter's ionosphere: Results from the first Galileo radio occultation experiment, *Geophysical Research Letters*, **24**, No. 17, 2107-2110, 1997.
- Høeg, P., A. Hauchecorne, G. Kirchengast, S. Syndergaard, B. Belloul, R. Leitinger and W. Rothleitner, Derivation of Atmospheric Properties using Radio Occultation Technique, *DMI Scientific Report 95-4*, 210 pp., Danish Meteorological Institute, Copenhagen, Denmark, 1996.
- Karaye, E. T. and D. P. Hinson, Sub-Fresnel-Scale Vertical Resolution in Atmospheric Profiles from Radio Occultation, *Radio Sci.*, **32**, No. 2, 411-423, 1997.
- Kursinski, E. R., G. A. Hajj, W. I. Bertinger, S. S. Leroy, T. K. Meehan, L. J. Romans, J. T. Schofield, D. J. McCleese, W. G. Melbourne, C. L. Thornton, T. P. Yunck, J. R. Eyre and R. N. Nagatani, Initial results of radio occultation observations of Earth's atmosphere using the Global Positioning System (GPS), *Science* **271**, 1107-1109, 1996.
- Marouf, E.A., G.L. Tyler and P.A. Rosen, Profiling Saturn's Rings by Radio Occultation, *ICARUS* **68**, 120-166, 1986.
- Melbourne, W.G., E.S. Davis, C.B. Duncan, G.A. Hajj, K.R. Hardy, E.R. Kursinski, T.K. Meehan, L.E. Young and T.P. Yunck, The Application of Spaceborne GPS to Atmospheric Limb Sounding and Global Change Monitoring, *JPL-publication 94-18*, 138 pp., JPL, Pasadena, CA, USA, 1994.
- Mortensen, M.D. and P. Høeg, Fresnel diffraction theory for enhanced vertical resolution in atmospheric profiling using GPS, submitted to *Geophysical Research Letters*, 1998.
- Ware, R., M. Exner, D. Feng, M. Gorbunov, K. Hardy, B. Herman, Y. Kuo, T. Meehan, W. Melbourne, C. Rocken, W. Screiner, S. Sokolovskiy, F. Solheim, X. Zou, R. Anthes, S. Businger and K. Trenberth, GPS Sounding of the Atmosphere from Low Earth Orbit: Preliminary Results, *Bull. Amer. Meteor. Soc.* **77**, 19-40, 1996.

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